Zirui (Ken) Deng

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EDUCATION

Washington University in St. Louis **St. Louis, Missouri**
PhD Candidate in Computer Science August 2021 – December 2026
· Research interest: theoretical and applied machine learning, information theory, privacy, and quantitative modeling

Emory University **Atlanta, Georgia**
B.S. in Applied Mathematics and Statistics & B.S. in Computer Science (Major GPA: 3.92/4.0) August 2017 – May 2021
· Coursework: Numerical Analysis, Calculus, Linear Algebra, Optimization, Partial Differential Equations, Probability & Statistics, Data Structures and Algorithms, Database Systems, Systems Programming, Deep Learning, Machine Learning

RESEARCH EXPERIENCE

Privacy Certification for ML Models via PMC **St. Louis, Missouri**
Washington University in St. Louis August 2025 – Present
· Designing a unified privacy auditing framework leveraging Pointwise Maximum Cost (**PMC**), enabling quantification of membership inference risk in **LLM**, **Transformer**, and **CNN**s, achieving 18% **AUPRC** improvement.
· Developing scalable estimation pipelines (**Python**, **PyTorch**, **Pandas**) linking empirical attack metrics (**TPR**/**FPR**) to theoretical guarantees, supporting production-grade ML model monitoring and reducing evaluation latency by 30%.
· Analyzing adversarial strategies across models (**SVMs**, **generative models**), deriving optimal inference policies and improving robustness of ML systems under distribution shift, reducing worst-case attack success by 20% (**AUROC**).

Efficient Vector Symbolic Architectures via Histogram Recovery **St. Louis, Missouri**
Washington University in St. Louis February 2025 – August 2025
· Built a neurosymbolic representation framework (**VSA**) for compositional learning, enabling efficient encoding and structured data retrieval across **Transformer** and **RNN**s (**LSTM**, **GRU**), achieving 30% improvement in retrieval accuracy.
· Deployed polynomial-time decoding algorithms (histogram recovery, lattice decoding) to replace prior NP-hard methods, boosting 10x speedup and enabling scalable inference in high-dimensional ML representations.
· Implemented end-to-end automated evaluation pipelines (**Python**, **PyTorch**, **CI/CD**) to benchmark noise robustness and retrieval accuracy, supporting memory-efficient representation learning in large-scale deployments.
(Link: https://ieeexplore.ieee.org/abstract/document/11654060)

Private Inference in Quantized Models **St. Louis, Missouri**
Washington University in St. Louis September 2022 – May 2025
· Constructed privacy-preserving ML inference protocols for **quantized neural networks** (**Regression**s, **CNN**s, etc.) using **Pandas**, **PyTorch**, and **scikit-learn**, enabling secure inner-product evaluation with <1% accuracy degradation.
· Developed & assessed information-theoretic privacy-utility tradeoff frameworks, reducing communication complexity by ~10x while maintaining strong guarantees against inference attacks in large-scale ML systems.
· Automated scalable evaluation pipelines (**Python**, **PyTorch**) to benchmark performance across high-dimensional datasets, improving attack resistance vs. cryptographic baselines and validating robustness through stress testing.
(Links: https://ieeexplore.ieee.org/abstract/document/10896693, https://ieeexplore.ieee.org/abstract/document/10206883)

Perfect Subset Privacy in Polynomial Computation **St. Louis, Missouri**
Washington University in St. Louis October 2023 – April 2025
· Designed distributed ML computation frameworks using **Reed-Muller** / **coded computing**, enabling privacy-preserving training and inference for **GNN**s (**LightGCN**) and large-scale data pipelines.
· Optimized & analyzed **NLP** encoding schemes with zero information leakage guarantees, achieving communication cost of lower orders of magnitude and improving scalability over naive distributed ML baselines.
· Established simulation pipelines (**Python**, **Pandas**) to evaluate robustness to system failures (e.g., stragglers), improving fault tolerance by ~25% for distributed ML workloads and large-scale data processing systems.
(Links: https://ieeexplore.ieee.org/document/10619107, https://ieeexplore.ieee.org/abstract/document/11677220)

Predicting Surgical Case Duration Using Neural ODEs **St. Louis, Missouri**
Washington University in St. Louis August 2021 – November 2021
· Designed & implemented a time-series prediction pipeline using **Python**, **PyTorch**, integrating **LSTM** and **Neural ODE** models to capture both discrete and continuous dynamics in heterogeneous clinical data.
· Developed robust data preprocessing workflows for multimodal datasets (static + temporal features), improving training stability and enabling scalable experimentation across **RNN/LSTM-based** architectures.

· Optimized numerical solvers and training procedures for stability and efficiency, reducing surgical duration prediction error by 15% **(CRPS)** and improving reliability in real-world healthcare forecasting tasks.

Undergraduate Honors thesis: Fair algorithms in machine learning **Atlanta, Georgia**
Emory University May 2020 – May 2021
· Analyzed algorithmic fairness in **ML classification and clustering** models, defining fairness via algorithmic bias in ML-based decisions on sensitive variables (e.g., gender, ethnicity, sexual orientation).
· Developed and evaluated optimization-based fairness constraints (e.g., parity regularization, constrained ERM), reducing statistical parity gap by up to 40% with <5% loss in predictive accuracy.
· Evaluated fairness-utility tradeoffs under distributional shifts, showing improved out-of-sample generalization with reduced disparity by ~30% and stable predictive performance (≤3% drop in accuracy) on held-out test sets.

Multi-variate Neural Ordinary Differential Equations for Time-series Modeling **Atlanta, Georgia**
Remote from Emory University May 2020 – July 2020
· Designed and implemented deep learning models (**CNN**, **LSTM**, **Neural ODE**) for time-series medical data, capturing discrete and continuous dynamics and reducing forecasting error by ~12% (**MSE**) relative to baseline **RNN** models.
· Developed data augmentation and contrastive learning strategies to improve representation quality, achieving 8-10% improvement in downstream prediction accuracy and matching state-of-the-art sequence modeling benchmarks.
· Optimized training pipelines for stability and scalability (batching, learning rate scheduling, numerical stability), reducing training time by ~40% and enabling efficient experimentation across large-scale time-series datasets.

INTERNSHIPS

Data Analyst Internship **Nanjing, China**
China Telecom May 2019 – June 2019
· Developed automated data pipelines using **Python, SQL, Pandas**, enabling large-scale web scraping (10K+ records/day) and improving data collection efficiency by ~60% for financial analysis workflows.
· Implemented validation and cleaning scripts (**Python, NumPy**) to ensure data integrity, reducing data inconsistencies by ~35% and lowering manual processing time by ~50%.
· Contributed to a financial robotics project by building modular data processing components, supporting near real-time ingestion and improving downstream analytics latency by ~30%.

Investment Banking Analysis Internship **Shanghai, China**
Ten Billion Research Laboratory in Shanghai May 2018 – June 2018
· Conducted quantitative market research on blockchain technologies using **Python, Excel**, analyzing 50+ projects/datasets and identifying key industry trends through SWOT and time-series analysis.
· Developed forecasting insights from structured and unstructured data, improving prediction accuracy of short-term market trends by ~20% compared to baseline heuristic models.
· Presented data-driven findings in cross-functional meetings, delivering actionable investment insights that informed strategy discussions and improved decision turnaround time by ~25%.

ADDITIONAL INFORMATION

Languages: Fluent in English and Mandarin Chinese
Interests: Poker; Weiqi (amateur 5-dan, national second-level athlete)